AMD DEVELOPER HACKATHON · ACT III

Machine Sound Monitor

Listen before it breaks.

Track 1 · Intelligent Industry

AI acoustic monitoring

THE PROBLEM

Machines rarely fail in silence.
Nobody is listening.

Weeks before a fan, pump or motor breaks, it starts to sound different. The change is usually noticed only when it is loud enough to bother someone.

1

Can't be everywhere

A maintenance team cannot stand next to every machine.

2

Noticed too late

The early change is quiet. It is heard once it is loud.

3

Downtime

By then it means an unplanned stop and an expensive repair.



THE SOLUTION

A device already on site becomes a listener.

Put it next to the machine. It learns **that machine's own normal sound**, then raises one clear alert when unusual sound persists.

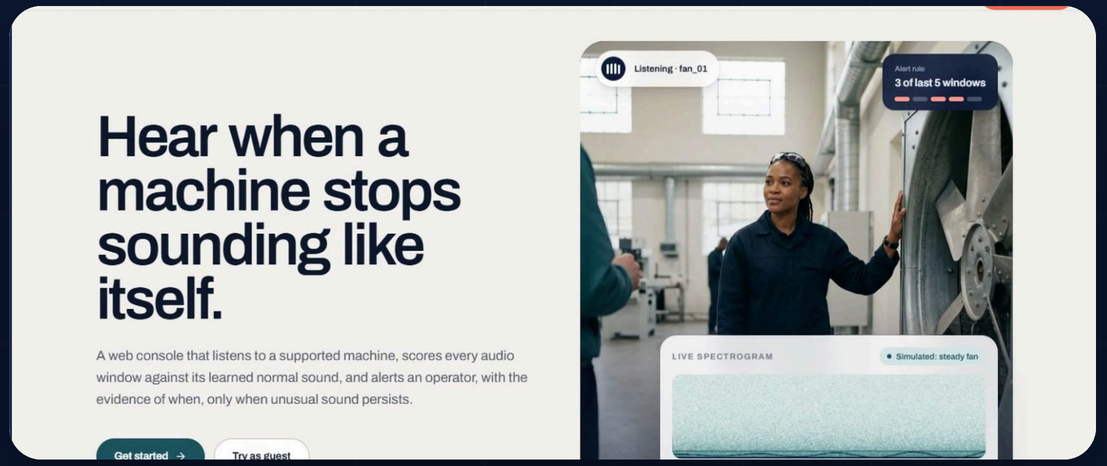
Phone or laptop browser

Works today: open the listener link, allow the microphone.

IoT device with a sound sensor

Sends its audio windows to the same API with its own device key.

No special hardware | **No account needed to try**



HOW IT WORKS

From sound to one useful alert.

01

Listen

A 3-second window every second, resampled to 16 kHz.

02

Compare

Sound **character** vs the learned normal, not volume. A level change is a note, never an alert.

03

Ignore one-offs

A door slam is one window. Alerts need 3 of the last 5 windows.

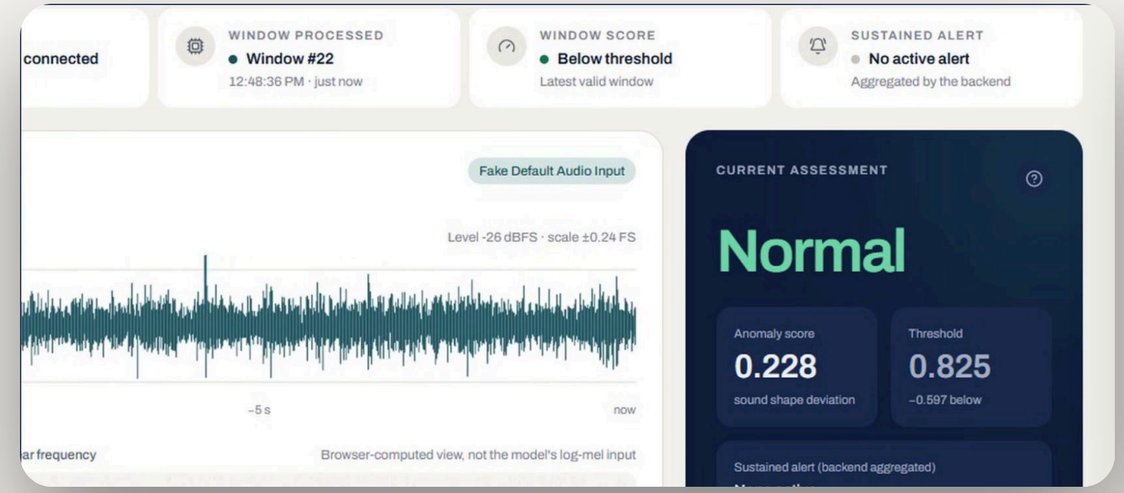
04

One alert

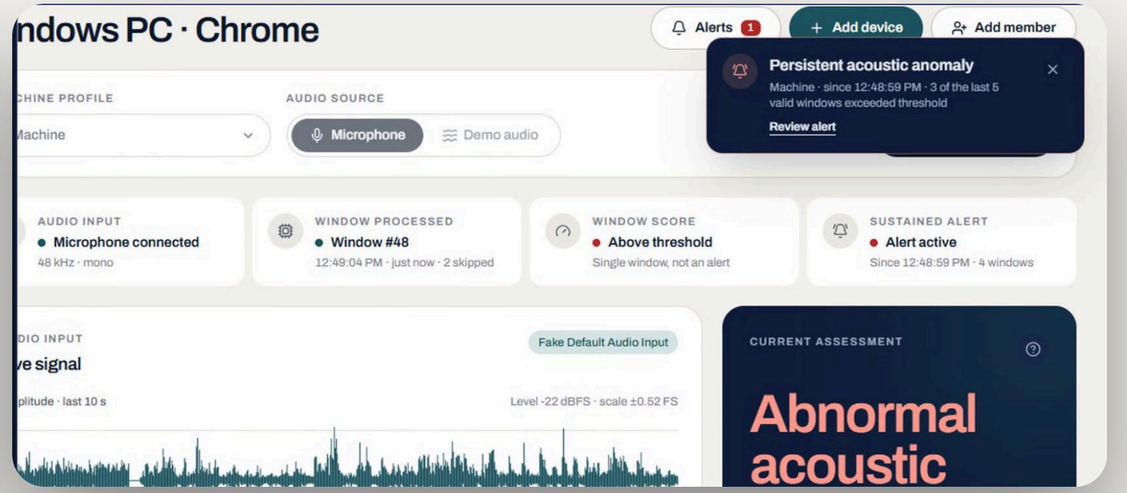
Email to the workspace members. Live on every dashboard. Quiet until it recovers.



Normal, until it isn't.

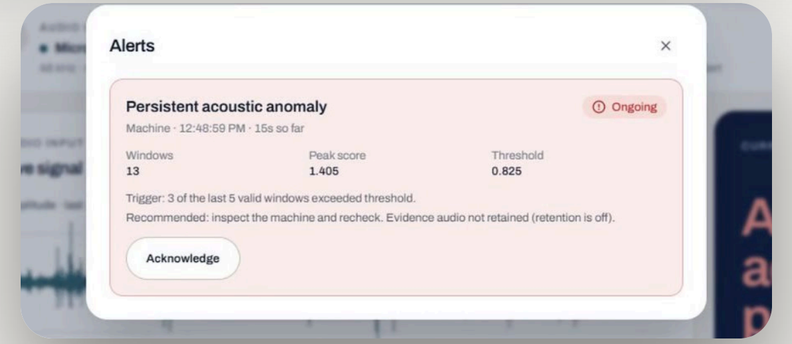


EVERY WINDOW UNDER THE LINE



THE FAN DEVELOPS A RATTLE → ONE ALERT

Each alert records **when it started, how many windows triggered it, and the peak score**, and recommends a person inspects and rechecks. Anyone on the dashboard can acknowledge it.



LIVE DEMO

Try it in under a minute.

1

Try as guest

Straight into a workspace, no account.

2

Add device

This computer's microphone, or a phone via its listener link.

3

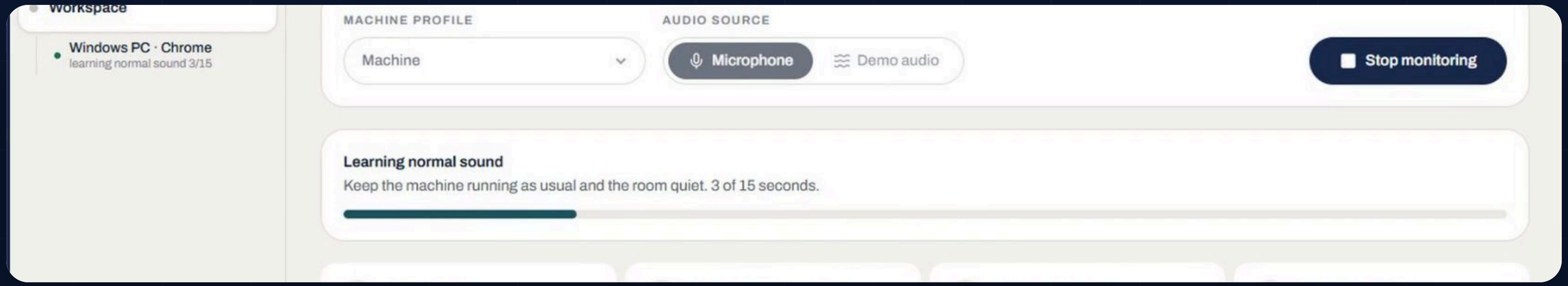
Learn

About a minute of normal sound (15 s in the demo), then locked.

4

Listen & alert

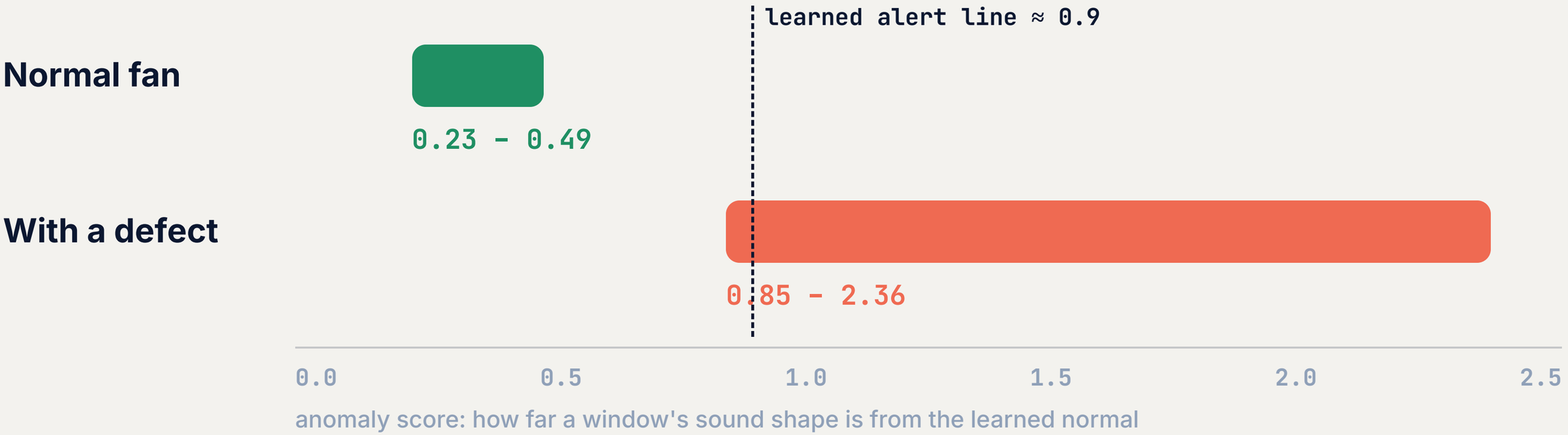
Scores live; a sustained change raises one alert and an email.



TESTED ON A REAL RECORDING

The defect stands clear of normal.

A phone recording of a desk fan, then the same fan with a rattle. On-site scorer, learned from that fan's own normal minute.



HOW WE BUILT IT

Architecture

Listening device

Phone or laptop browser (React, TypeScript, Web Audio API). IoT devices post to the same API.

WAV
device key

FastAPI backend

Validate → resample 16 kHz → score → 3-of-5 alert policy → notify. Live stream to dashboards (SSE).

The team

Email alerts via Resend. Dashboards on any screen. Acknowledge.

- **Raw audio never stored**
- Device keys stored as SHA-256 hashes
- Supabase sign-in + Postgres
- Workspaces with view / edit members
- Frontend on Vercel

HOW WE BUILT IT

The models, and AMD.

Two scorers behind one interface. The live demo uses the **on-site learner** (band-energy shape vs the learned minute). For industrial machines we trained **PyTorch autoencoders** on healthy log-mel audio only.

DCASE 2020 · FAN (HELD-OUT TEST)	ROC AUC
Dense autoencoder (ours)	0.646
PANNs CNN14 embedding + kNN	0.594
Conv autoencoder (ours)	0.503
Official DCASE baseline	≈ 0.66

BUILT FOR

AMD Instinct MI300X

The training and evaluation pipeline is packaged for AMD Developer Cloud: one script, `ml/run_amd.sh`, PyTorch on ROCm. Reports record the GPU and ROCm version.

PyTorch · ROCm

AMD Developer Cloud

Next: run the full training on the MI300X and serve the model from the AMD machine.

BUILT FOR REAL ROOMS

Honest about what it hears.

?

Says when it can't tell

"Unable to assess" when the signal is too weak, instead of guessing.

1×

Ignores one-off spikes

A door slam is one window. An alert needs 3 of the last 5.

dB

Character, not volume

A louder or quieter room is a note on the window, never an anomaly.

✓

One alert per incident

Then quiet until recovery, plus a 30 s cooldown. No alert floods.

◌

No audio kept

Only scores and events are stored. Raw audio is never saved.

!

Not a diagnosis

An early warning that asks a person to check. Not a certified safety system.

IMPACT & NEXT STEPS

Hear a fault earlier, with devices you already own.

Who it's for

Maintenance teams in plants, workshops and buildings with fans, pumps and motors that nobody watches all day. Start with one phone next to one machine, no installation project.

Roadmap

- Train and serve on AMD Instinct MI300X
- A small always-on IoT listener client
- Per-machine trained models, from each site's healthy audio
- A fleet view across many machines



Machine Sound Monitor

Listen before it **breaks.**

github.com/KieruGIT/AMD-ACT-III

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